Sahaj Bhadja

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Summary

Software engineer with 2+ years building scalable **microservices, APIs, and AI systems** using Python, Java, and C++; experienced in architecture, relational schema design, system testing, monitoring, debugging, and production reliability.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Architected scalable multi-tenant backend microservices using Python, FastAPI, PostgreSQL, and SQL Server, supporting 500+ users while reducing API and dashboard latency from 1.2 seconds to 350 milliseconds.
- Improved quality, performance, and monitoring for 5,000+ recurring monthly workflows through testing, validation, reconciliation, error handling, and production debugging, sustaining 99%+ reliability across multiple integrated applications.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Built resilient AI-powered backend microservice APIs using Python, Flask, and AWS, serving 10,000+ daily requests with concurrent processing, failure isolation, and stable performance for 150+ instructors.
- Managed feature priorities and deliverables from technical requirements through system testing and documentation, resolving production defects and reducing feedback turnaround 60% while maintaining 90%+ AI grading consistency.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built scalable machine-learning pipelines processing 100,000+ ride trajectory logs, improving long-horizon planning 18% through Python model development, data structures, extensive algorithm evaluation, and reproducible closed-loop simulation.
- Designed robust architecture combining road geometry, lane direction, and vehicle context, then validated system behavior across 10,000 scenarios, improving planning stability 15% and reducing off-road trajectories.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Improved production microservice APIs using object-oriented patterns, AWS Glue, and S3, reducing response times 30%, payload sizes 25%, technical debt 20%, and data-processing overhead across distributed systems.
- Triaged, debugged, and resolved critical cross-platform issues while maintaining 20+ unit and integration tests, coordinating global releases with Microsoft and GitHub engineers across Windows, Linux, and macOS.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Implemented eight independent production microservices for resource allocation, customer management, billing, permissions, and notifications, synchronizing two million records across locations through REST APIs, WebSockets, and Azure SQL.
- Established comprehensive monitoring with Sentry, Azure Monitor, and Kibana while automating repeatable system testing and safe staged deployments, achieving 95%+ build validation and reducing code-review turnaround 25%.

Skills

Languages: Python, Java, C, C++, JavaScript, TypeScript, SQL, Shell
Backend: FastAPI, Flask, Node.js, Express.js, REST APIs, microservices, distributed systems, object-oriented programming
Data: PostgreSQL, SQL Server, Azure SQL, MySQL, MongoDB, relational databases, NoSQL, schema design, query optimization
Cloud: AWS, S3, Glue, RDS, Azure, Docker, Kubernetes, CI/CD, scalable architecture
Quality: System testing, validation, unit testing, integration testing, debugging, issue triage, monitoring, Sentry, Kibana, Git, Jira

Projects

HMIS Live | *Microservices, TypeScript, Monitoring*

- Built a microservice-based healthcare platform with TypeScript, REST integrations, Sentry monitoring, and real-time notifications, supporting 100+ daily workflows while reducing record retrieval time 50% and administration 30%.

Chat2SQL Code | *Java, TypeScript, Schema Design*

- Extended a Java and TypeScript platform supporting 10+ relational and NoSQL databases, implementing schema exploration, query validation, structured execution, and cross-platform debugging within a unified developer experience.

CPU Scheduling Simulator Live | *C++, Algorithms, System Testing*

- Implemented six C++ scheduling algorithms with edge-case handling, interactive controls, and Gantt visualization, validating correctness across 1,000+ process combinations while reducing average turnaround time 25% and debugging 20%.